

Two-Stage, Bias-Audited Resume Screening with Explainable, Redaction-Aware Ranking

Project Group Members

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Abstract

Automated resume screening promises to save recruiter time, but two problems undercut its trustworthiness: hard requirements and graded fit are usually collapsed into a single opaque score, and language models are known to pick up on identity signals - name, gender markers, university prestige - that have nothing to do with fitness for a role. This project builds a Resume Screening system that keeps those as two separate, individually measurable stages rather than one number, and pairs its ranking with a measured, reproducible bias audit rather than a disclaimer bolted on after the fact. The system is advisory only: it never rejects a Candidate and never discards one from its output.

Given a Role and a Recruiter-approved Requirement Set, Screening tests every Candidate against each Requirement and returns a binary outcome with a citable Justification; a Candidate who fails is still returned, marked with the Requirement they missed, and a Candidate whose verdict never parses becomes an Unresolved Candidate rather than a silent rejection. Requirements are proposed by LLM extraction from the Job Description, but none enters the Requirement Set until a Recruiter approves it, so a hallucinated Requirement can never silently disqualify anyone. Every Candidate is passed through Screening and, if Qualified, through Ranking - there is deliberately no vector or embedding prefilter narrowing the pool first, since anything a prefilter discarded would never reach the metrics.

Ranking sees only a Redacted Resume: direct identity signals such as name, gender markers, and graduation year are stripped before a graded, non-binary Fit judgement is made, while a Contested Signal such as university is deliberately left in so its effect can be measured rather than engineered away. Counterfactual Sensitivity is reported as a first-class result rather than an afterthought - counterfactual Resume pairs differing in exactly one identity signal at a time, across name/gender, name/ethnicity, and university axes, are scored with redaction on and off, and the resulting shift in rank position and Fit is published as a measured delta.

The system's output, a Shortlist, is measured with rank metrics rather than classification accuracy. Relevance at scale is constructed as Proxy Relevance from a category-labelled resume corpus and checked against a small, hand-labelled Gold Set judged against a written rubric, because the coarse automatic label can be wrong in either direction and a divergence between the two is treated as a finding rather than a bug. The result is a screening pipeline whose rankings are explainable by construction, whose fairness claims rest on a repeatable audit rather than a single anecdote, and whose every failure mode - a missed Requirement, an unparsed verdict, a proxy-versus-human disagreement - is a published number rather than a hidden one.

Datasets

Dataset	Source
Resume corpus (ahmedheakl/resume-atlas)	Resume text and category labels across 43 job categories, used as the Applicant Pool and as the source of Proxy Relevance.
Job postings corpus (NextGig-Rocks/global-job-postings-multi-ats)	Real Job Descriptions matched by script to each corpus category, forming the checked-in Evaluation Roles; never derived from the resumes themselves.
Gold Set	Roughly 150 Role-and-Resume pairs judged by hand against a written rubric, sampled deterministically across same-, adjacent-, and distant-category cases, used to check Proxy Relevance rather than replace it.
Synthetic counterfactual pairs	Generated in-pipeline by altering one identity signal at a time (name/gender, name/ethnicity, university) in a Redacted Resume, used to measure Counterfactual Sensitivity.

Approach

